

2024 Calculus II Additional Exercise (5)

1. Exercise 2 in page 32 from Calculus Supplement Homework.
2. Let C be a parametric curve given by the equation $x = a(t - \sin t)$, $y = a(1 - \cos t)$, $t \in [0, 2\pi]$.
 - (a) Find the area enclosed by the curve C and the x -axis.
 - (b) Find the length of the curve C .
 - (c) Find the area of the surface generated by revolving the curve C about x -axis.
 - (d) Find the centroid of the curve C .
3. Let C be given by the equation $r^2 = a^2 \cos 2\theta$, $\theta \in [-\frac{\pi}{4}, \frac{\pi}{4}]$.
 - (a) Graph the curve C in the Cartesian xy -plane.
 - (b) Find the area enclosed by the curve C .
4. Exercise 4 in page 32 from Calculus Supplement Homework.
5. Exercise 12 in page 33 from Calculus Supplement Homework.